

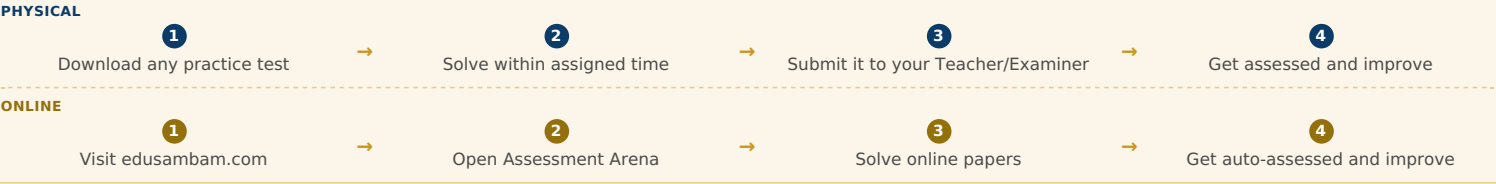
EDUSAMBAM ASSESSMENT SERIES

Number Systems

PRINTABLE PRACTICE WORKSHEET

EDUSAMBAM, working on building your Foundational Concepts. Visit edusambam.com for Auto-graded Self-assessment Experience.

YOUR PHYSICAL & ONLINE SELF-ASSESSMENT ROADMAP



Candidate Name:		Roll No.:	
Class / Grade:		Date:	
School / Institution:		Test Centre:	

Subject	Mathematics (Maths)	Topic	Number Systems	Total Marks	100
Total Questions	45	Time Allowed	1 Hour (60 minutes)	Version	A

GENERAL INSTRUCTIONS

- Read each question carefully before answering.
- For Section A and Section B, mark one answer only.
- For Sections C and E, write your answer clearly in the space provided.
- For Section D, write the correct letter from Column B against each item in Column A.
- Do not use correction fluid. Keep all answers clear and legible.
- This worksheet is prepared from the EDUSAMBAM practice assessment supplied for this paper.

PAPER PATTERN

Section	Questions	Marks Each	Total
Section A — Multiple Choice	15	2	30
Section B — True / False	10	2	20
Section C — Fill in the Blank	10	2	20
Section D — Column Matching	5 pairs	2	10
Section E — Short Answer	5	4	20
Total	45	—	100

Section A — Multiple Choice

30 marks

1. The place value of a digit depends on:

A. Its position in the number

B. Its color

C. The size of the page

D. Nothing

Answer: ☐A ☐B ☐C ☐D

Basic • 2 pts

2. The face value of a digit is:

A. The digit itself, regardless of its position

B. Always zero

C. The digit multiplied by 100

D. The number of digits in the number

Answer: ☐A ☐B ☐C ☐D

Basic • 2 pts

3. In the number present, natural numbers start counting from:

A. Negative infinity

B. 1 (or 0 depending on convention), and go upward

C. Only even numbers

D. Only fractions

Answer: ☐A ☐B ☐C ☐D

Basic • 2 pts

4. Whole numbers include:

A. Natural numbers plus zero

B. Only negative numbers

C. Only fractions

D. Only decimals

Answer: ☐A ☐B ☐C ☐D

Basic • 2 pts

5. Integers include:

A. Only positive numbers

B. Positive numbers, negative numbers, and zero

C. Only fractions

D. Only decimals

Answer: ☐A ☐B ☐C ☐D

Intermediate • 2 pts

6. A rational number can be expressed as:
A. A fraction of two integers, with a non-zero denominator
C. Never as a fraction
Answer: ☐A ☐B ☐C ☐D

B. Only a whole number
D. Only a negative number

Intermediate • 2 pts

7. Which of these is an example of a rational number?
A. 3/4
C. Pi (π)
Answer: ☐A ☐B ☐C ☐D

B. The square root of 2 (an irrational number)
D. None of these

Intermediate • 2 pts

8. Real numbers include:
A. Only whole numbers
C. Only negative numbers
Answer: ☐A ☐B ☐C ☐D

B. Both rational and irrational numbers
D. Only fractions

Intermediate • 2 pts

9. Roman numerals use letters such as:
A. I, V, X, L, C, D, M
C. 1, 2, 3, 4
Answer: ☐A ☐B ☐C ☐D

B. A, B, C, D
D. Greek letters only

Intermediate • 2 pts

10. In Roman numerals, the letter "X" represents:
A. 5
C. 50
Answer: ☐A ☐B ☐C ☐D

B. 10
D. 100

Intermediate • 2 pts

11. The binary number system uses only the digits:
A. 0 and 1
C. 1 through 5
Answer: ☐A ☐B ☐C ☐D

B. 0 through 9
D. A through F

Intermediate • 2 pts

12. The decimal number system is based on:
A. Base 2
C. Base 16
Answer: ☐A ☐B ☐C ☐D

B. Base 10
D. Base 8

Advanced • 2 pts

13. The octal number system is based on:
A. Base 2
C. Base 10
Answer: ☐A ☐B ☐C ☐D

B. Base 8
D. Base 16

Advanced • 2 pts

14. Which number base system is commonly used in computing at the lowest level?
A. Binary (base 2)
C. Base 5
Answer: ☐A ☐B ☐C ☐D

B. Roman numerals
D. Base 7

Advanced • 2 pts

15. In the International number system, place values include ones, tens, hundreds, and:
A. Thousands
C. Only tens
Answer: ☐A ☐B ☐C ☐D

B. Only ones
D. None of these

Advanced • 2 pts

Section B — True / False

20 marks

16. Place value depends on where a digit is positioned in a number.
A. True ☐ **B.** False ☐

Basic • 2 pts

17. Face value of a digit changes depending on its position in the number.
A. True ☐ **B.** False ☐

Basic • 2 pts

18. Natural numbers are used for basic counting.
A. True ☐ **B.** False ☐

Basic • 2 pts

19. Whole numbers include zero along with all natural numbers.
A. True ☐ **B.** False ☐

Basic • 2 pts

20. Integers can be negative, zero, or positive.
A. True ☐ **B.** False ☐

Intermediate • 2 pts

21. Every rational number can be written as a fraction of two integers.
A. True ☐ **B.** False ☐

Intermediate • 2 pts

22. Irrational numbers, like pi, can be written as an exact simple fraction.
A. True ☐ **B.** False ☐

23. Real numbers include both rational and irrational numbers.
A. True ☐ **B.** False ☐

24. Roman numerals use the standard Hindu-Arabic digits 0 through 9.
A. True ☐ **B.** False ☐

25. The binary number system uses only two digits: 0 and 1.
A. True ☐ **B.** False ☐

Intermediate • 2 pts

Intermediate • 2 pts

Intermediate • 2 pts

Intermediate • 2 pts

Section C — Fill in the Blank

20 marks

26. The value of a digit based on its position in a number is called its ____ value.
Answer: _____

27. The value of a digit itself, regardless of position, is called its ____ value.
Answer: _____

28. Numbers used for basic counting, like 1, 2, 3, are called ____ numbers.
Answer: _____

29. Natural numbers plus zero are called ____ numbers.
Answer: _____

30. Numbers that include positive numbers, negative numbers, and zero are called ____.
Answer: _____

31. A number that can be written as a fraction of two integers is called a ____ number.
Answer: _____

32. A number that cannot be written as an exact fraction, like pi, is called an ____ number.
Answer: _____

33. The ancient number system that uses letters like I, V, and X is called ____ numerals.
Answer: _____

34. The number system that uses only 0 and 1 is called the ____ system.
Answer: _____

35. The number system based on 10 digits (0-9) that we use every day is called the ____ system.
Answer: _____

Basic • 2 pts

Basic • 2 pts

Basic • 2 pts

Basic • 2 pts

Intermediate • 2 pts

Intermediate • 2 pts

Intermediate • 2 pts

Intermediate • 2 pts

Intermediate • 2 pts

Intermediate • 2 pts

Section D — Column Matching

10 marks

36-40. Match each term to its correct definition.

Column A	Answer	Column B
Place Value	____	A. A number that cannot be written as an exact simple fraction
Rational Number	____	B. An ancient number system using letters like I, V, and X
Irrational Number	____	C. A number system that uses only the digits 0 and 1
Roman Numerals	____	D. The value of a digit based on its position in a number
Binary System	____	E. A number that can be expressed as a fraction of two integers

Section E — Short Answer

20 marks

41. Explain the difference between place value and face value.

42. What is the difference between whole numbers and natural numbers?

Advanced • 4 pts

Advanced • 4 pts

43. What makes a number a rational number?

Advanced · 4 pts

44. Give one example of a real-life or historical use of Roman numerals.

Advanced · 4 pts

45. Why is the binary number system important in computing?

Advanced · 4 pts

Candidate Signature: _____ **Invigilator Signature:** _____

FOR OFFICE USE ONLY

Marks Obtained: _____ / 100

Checked By: _____

Result / Grade: _____

Date Checked: _____

EDUSAMBAM ASSESSMENT SERIES

Number Systems — Answer Key

Teacher reference: equivalent grammatically correct wording is acceptable for open-ended items when the meaning is preserved.

Section A — Multiple Choice

- | | |
|------|-------|
| 1. A | 9. A |
| 2. A | 10. B |
| 3. B | 11. A |
| 4. A | 12. B |
| 5. B | 13. B |
| 6. A | 14. A |
| 7. A | 15. A |
| 8. B | |

Section B — True / False

- | | |
|-----------|-----------|
| 16. True | 21. True |
| 17. False | 22. False |
| 18. True | 23. True |
| 19. True | 24. False |
| 20. True | 25. True |

Section C — Fill in the Blank

- | | |
|--------------|----------------|
| 26. place | 31. rational |
| 27. face | 32. irrational |
| 28. natural | 33. roman |
| 29. whole | 34. binary |
| 30. integers | 35. decimal |

Section D — Column Matching

- | | |
|---------------------------|------------------------|
| 36. Place Value — D | 39. Roman Numerals — B |
| 37. Rational Number — E | 40. Binary System — C |
| 38. Irrational Number — A | |

Section E — Short Answer (Model Answers)

- 41.** position; itself; regardless; location
- 42.** zero; counting; start
- 43.** fraction; integer; ratio; expressed
- 44.** clock; book; chapter; monument; movie; king; queen
- 45.** computer; digital; 0 and 1; electronic; processor